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Otoscopic diagnosis of middle ear effusion in acute and non-acute otitis media. I. The value of different otoscopic findings

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Abstract

To determine the value of different pneumotoscopic findings in diagnosing the middle ear effusion (MEE) of acute (AOM) and non-acute otitis media, 11,804 ear-related visits of 2911 unselected children at ages 0.5–2.5 years were analysed. About half of these were examined by an otolaryngologist in one, and half by a pediatrician in another, urban area. Myringotomy was always performed when MEE was suspected, and it confirmed the presence of MEE in 85% (otolaryngologist) and 82% (pediatrician) of altogether 5462 acute and in 69% (both doctors) of 1092 non-acute cases suspected. Redness of the tympanic membrane (TM) was found in only 18% and 27% of the visits with AOM, and it predicted MEE with only 60% and 51% probability, if seen in acute visits. Cloudiness of the TM was noticed in 81% and 67% of the visits with AOM; its specificity and the other calculated variables were good in regard to the diagnosing of MEE, especially in acute cases in both groups. Distinctly impaired mobility of the TM was of about the same diagnostic value, but its position reliably indicated MEE only when bulging. In

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AOM the colour or mobility of the TM was normal very rarely, but the position was normal in a third of the cases. Thus, although there were differences in the incidences of different otoscopic findings in the two study groups, the diagnostic value of certain pneumatic otoscopic findings, especially cloudiness and distinct hypomobility of the TM, seemed to be good in both groups.

Introduction

For the clinical diagnosis of acute otitis media (AOM), in addition to acute ear-related symptoms, signs of middle ear effusion (MEE) must also be present [2,8]. The presence or absence of MEE is usually assessed by (pneumatic) otoscopy, but in non-acute cases also tympanometrically or by hearing tests. The otoscopic criteria of AOM (and its MEE) vary greatly among clinicians as well as in different studies [6]. Redness of the tympanic membrane (TM) [11], its bulging [10] or impaired mobility [2] are all suggested to be indicative of AOM. However, opinions on the consistency of these findings in AOM differ [2,7,13,15], and they can all also result from causes other than middle ear suppuration. So, in AOM the real diagnostic value of pneumatic otoscopy and different otoscopic findings is unclear. In non-acute otitis media pneumatic otoscopy (with certain sets of findings) has been reported to detect 81–94% of ears with MEE (sensitivity) and 74–78% of ears without MEE (specificity) [4,12].

Tympanometry has mainly been used to screen and diagnose non-acute MEE. In myringotomy-controlled studies, however, the accuracy of tympanometry to diagnose MEE varied greatly, with sensitivity ranging from 65 to 99% and specificity from 45 to 100% [3,4]. In two studies on AOM, tympanometry was reported to detect MEE with unacceptably low reliability [14,17].

Among other alternatives to detect MEE, hearing tests, either with voice, tuning forks or audiometrically, are not feasible in AOM of small children. Sometimes the conductive hearing loss caused by MEE may also be too slight to be detectable even in older children [8]. Furthermore, studies of the usefulness of acoustic reflectometry in diagnosing MEE have given contradictory results even in non-acute otitis media [1,16].

Thus, it seems that the role of pneumatic otoscopy is still of primary importance in diagnosing MEE, especially in AOM. In this study we tried to determine the value of different otoscopic findings in detecting the MEE of acute and non-acute otitis media in two large unselected groups of children of the same ages, but living in different geographical areas. Each group was examined by a single study doctor, an otolaryngologist or a pediatrician. Different otoscopic findings were compared with those of myringotomy. Also interobserver and intergroup variations in otoscopic findings and their effects on the clinical diagnostics of otitis media were studied.

Materials and Methods

The patient material comprised 2911 children who were followed for 1–2 years for otitis episodes at age 0.5–2.5 years in two urban areas (the cities of Tampere and Oulu) in Finland. They formed 67.0% of the total number of children of their ages in the areas. During the follow-up, whenever the children had any ear-related problems or even a suspicion of them, they were brought to the Outpatient Clinics of Otolaryngology at the Tampere or Pediatrics at the Oulu University Central Hospital. In each study clinic the children were examined and treated by a single study doctor, an otolaryngologist in Tampere and a pediatrician in Oulu. Both doctors were well-trained and experienced in diagnosing and treating otitis media in children. Pneumatic otoscopy was always performed and the TM findings, such as colour, position, mobility or possible perforation or the existence of a tympanostomy tube, were carefully recorded. Myringotomy and aspiration were always performed when MEE was even suspected in pneumatic otoscopy. A child was diagnosed as having AOM if MEE was found and the child had any of the following symptoms: fever, earache, irritability, ear rubbing or tugging, simultaneous other acute respiratory symptoms, vomiting or diarrhea. All the visits without the above symptoms were called non-acute. In addition to myringotomy and aspiration, AOM attacks were treated with antimicrobials, primarily penicillin V. Children were re-examined at 2–3 week intervals in the study clinic until the ears were completely healed. Myringotomy was always repeated if MEE was suspected.

Altogether 11,804 visits (53.2% in boys) were available for analysis, visits with a tympanostomy tube in one or both ears (1324) and visits with incomplete data (990) excluded. In Tampere there were 5949 visits in 1688 children (Group I), and in Oulu 5855 visits in 1223 children (Group II). The age distribution during the visits was similar in both groups (Table I).

Of all the visits 75.1% were due to acute symptoms (Table II). Myringotomy was performed in 61.7% of acute visits and in 37.1% of the visits without acute symptoms, and MEE was obtained in 83.5% and 69.0% of these, respectively. TM

TABLE I

Age of children during ear-related visits

Age (months)	Group I		Group II	
	No	%	No	%
6–11	2183	36.7	2106	36.0
12–17	2458	41.3	2221	37.9
18–23	1087	18.3	1172	20.0
24–29	214	3.6	344	5.9
30–51	7	0.1	12	0.2
Total	5949		5855	

Groups I and II = groups of children examined by two different doctors in two different areas; see Materials and Methods.

TABLE II

Ear-related visits in different clinical categories

<i>Type of visit</i>	<i>Group I (1688)</i>		<i>Group II (1223)</i>		<i>Total (2911)</i>	
	<i>No</i>	<i>%</i>	<i>No</i>	<i>%</i>	<i>No</i>	<i>%</i>
All	5949		5855		11804	
with acute symptoms	4469	75.1	4390	75.0	8859	75.1
and myringotomy	3098	69.3	2364	53.8	5462	61.7
with MEE (AOM)	2630	58.8	1933	44.0	4563	51.5
without MEE	468	10.5	431	9.8	899	10.1
perforation	186	4.2	186	4.2	372	4.2
without acute symptoms	1480	24.9	1465	25.0	2945	24.9
and myringotomy	593	40.1	499	34.1	1092	37.1
with MEE (= non-acute OM)	408	27.6	345	23.5	753	25.6
without MEE	185	12.5	154	10.5	339	11.5
perforation	28	1.9	69	4.7	97	3.3

Group = group of children; see Materials and Methods.

Number o. children followed-up is given in parentheses.

perforation was found in one or both ears in 4.2% of acute visits and in 3.3% of non-acute visits; these visits were excluded from analyses of TM mobility or position. In each visit the ear with the more severe TM findings was recorded. The order of severity of the findings was for the colour: haemorrhagic, strongly red, moderately red, cloudy (dull), slightly red, and normal; for the position: bulging, retracted and normal; and for the mobility: distinctly impaired, slightly impaired and normal. When relating the TM and middle ear (MEE) findings in each analysis, only visits with complete otoscopic and other necessary data were included. Thus the total number of visits (cases) somewhat varied from one analysis to another.

The diagnostic value of different otoscopic findings in diagnosing MEE was evaluated by certain variables calculated as percentages from the original figures: (1) sensitivity, Se (the presence of a finding in the presence of MEE); (2) specificity, Sp (the absence of a finding in the absence of MEE); (3) predictive value (the presence of MEE in the presence of a finding, or the absence of MEE in the absence of a finding); (4) false-positive cases (the absence of MEE in the presence of a finding); (5) false-negative cases (the presence of MEE in the absence of a finding); and (6) efficiency (the presence of a finding with MEE and the absence of the finding without MEE in all the cases) [5]. To characterize the discriminatory ability of the otoscopic findings we also determined (7) the likelihood ratio of a positive test result ($LR+$) and (8) the likelihood ratio of a negative test result ($LR-$) for each finding as derivatives of sensitivity and specificity. $LR+ = Se/(1 - Sp)$, and it tells how many times greater is the probability of a finding to be present in a visit with MEE as compared to visits without MEE. $LR- = (1 - Se)/Sp$, and gives the relative frequency of the absence of a finding in visits with MEE as compared to visits without MEE.

Because the diagnostic value of a finding is also dependent on the frequency of MEE in the study population, we also analyzed the predictive values of the findings by using (9) the odds. From the total incidence figures of MEE in each group of visits (F), we calculated the corresponding prior odds (OPr; odds before otoscopy) and posterior odds (OPo; odds for an otoscopic finding) for the presence of MEE as follows:

$$\text{OPr} = F/(1 - F); \quad \text{OPo} = \text{OPr} \left(\frac{\text{Se}}{1 - \text{Sp}} \right) = \text{OPr}(\text{LR} +).$$

The data were processed and calculated in the Tampere University Computer Centre with the SURVO and SURDA data processing systems. Statistical analyses were done using the chi-square test.

Results

Myringotomy confirmed AOM in 58.8% of all the acute visits in Tampere and in 44.0% in Oulu. However, in acute visits myringotomy yielded MEE, when done, with about the same frequency, 84.9% and 81.8%, and also the total incidence rates of AOM per 100 children were almost the same, 1.56 and 1.58, in the two groups of children in the two areas. During non-acute visits MEE was found by myringotomy (= non-acute otitis media) in about one-quarter of all the cases, and in 68.8% and 69.1% of the cases myringotomized in the two groups (Table II).

The occurrence of different otoscopic findings in clinically different ear-related visits (sensitivity) is presented in Table III. In visits with acute symptoms the TM was seen as red in 16.7% (Tampere, Group I) and 20.2% (Oulu, Group II); the percentages did not vary much whether the MEE was present (AOM) or not. In non-acute visits the redness of the TM was recorded in 5.1% and 8.8%, again with about the same frequency in visits with or without MEE. In visits with MEE, regardless of whether they were acute or not, on the other hand, cloudiness of the TM was a common finding, seen in 92.9–67.1%. Bulging of the TM was found in 61.2% and 41.3% of the acute visits with MEE. Like cloudiness, it was a rare finding if there was no MEE. Retraction of the TM was a rather rare finding in all subgroups of visits. If there was MEE, the mobility of the TM was almost always impaired. This finding was also recorded in 9.6–28.9% of the cases without acute or non-acute MEE. In visits with AOM the TMs were of normal colour in 1.1% and 6.0%, mobility in 1.7% and 5.6%, but position in 32.3% and 39.3% in the two groups.

The probability of different otoscopic findings to indicate the presence of MEE (predictive value) showed variations (Table IV). If there were acute ear-related symptoms, MEE (AOM) was found in 59.6% and 51.4% of the cases with TM redness. If redness was only slight, the probability to find MEE was very poor, increasing to 65–70% level in acute cases with more distinct TM redness. If the TM was cloudy, bulging or showed impaired mobility, and there were acute ear-related symptoms, MEE (AOM) was found usually in 80–95% frequency, somewhat varying

TABLE III

Tympanic membrane (TM) findings in visits of different clinical categories

TM-finding	Visits with acute symptoms				Visits without acute symptoms			
	MEE present (= AOM)		MEE absent		MEE present (= non-acute OM)		MEE absent	
	GI (%)	GII (%)	GI (%)	GII (%)	GI (%)	GII (%)	GI (%)	GII (%)
Colour								
red	17.5	26.8	15.5	16.0	4.7	12.2	5.2	8.1
distinctly red *	13.6	24.3	7.7	8.0	3.4	11.3	2.8	3.5
haemorrhagic	0.5	7.4	0.2	2.8	0.3	5.7	0	1.7
strongly red	5.7	6.9	1.0	2.0	0.8	2.8	0.7	0.5
moderately red	7.4	10.0	6.5	3.2	2.3	2.8	2.1	1.3
slightly red	3.9	2.5	7.8	8.0	1.3	0.9	2.4	4.6
cloudy	81.4	67.1	4.7	10.5	92.9	68.9	1.8	12.4
normal	1.1	6.0	79.8	73.5	2.5	18.9	93.0	79.5
Position								
bulging	61.2	41.3	3.2	3.1	45.2	18.4	0.8	0.8
retracted	6.5	19.4	9.4	12.1	10.0	32.1	5.1	9.6
normal	32.3	39.3	87.3	84.8	44.7	49.5	94.1	89.6
Mobility								
impaired	98.3	94.3	20.6	28.0	98.7	93.7	9.6	28.9
distinctly	82.6	84.0	6.9	14.6	76.3	84.3	2.6	14.2
slightly	15.7	10.3	13.7	13.4	22.4	9.4	7.0	14.7
normal	1.7	5.6	79.4	71.9	1.3	6.3	90.4	71.2

G = Group of children; see Materials and Methods.

* Haemorrhagic, strongly or moderately red.

TABLE IV

Middle ear effusion (MEE) in visits with different tympanic membrane (TM) findings (predictive value)

TM-finding	MEE present in visits			
	with acute symptoms		without acute symptoms	
	G I (%)	G II (%)	G I (%)	G II (%)
Colour				
red	59.6	51.4	24.0	23.0
distinctly red *	69.8	65.6	29.5	39.3
haemorrhagic	81.3	62.9	100.0	40.0
strongly red	87.7	68.1	27.2	54.5
moderately red	59.8	66.0	28.1	30.0
slightly red	39.4	16.7	16.1	3.8
cloudy	95.7	80.0	95.1	52.3
normal	1.7	4.9	1.0	4.5
Position				
bulging	96.0	89.0	95.7	81.4
retracted	46.8	49.6	41.9	38.6
normal	32.1	22.2	14.8	9.4
Mobility				
impaired	86.0	68.1	79.0	40.8
distinctly	94.0	78.5	91.5	55.8
slightly	59.7	32.8	54.0	12.0
normal	2.7	4.8	0.5	1.8

G = group of children; see Materials and Methods.

* Haemorrhagic, strongly or moderately red.

depending on which one of these findings was studied (Table IV). On the other hand, in cases with normal colour of the TM the MEE was absent in 95.1–99.0%, depending on the group studied, and in cases with normal TM mobility in 95.2–99.5%. Normal position of the TM predicted the absence of MEE in acute cases, however, with only 67.9% and 77.8% probability; in non-acute visits the probability was 85.1% and 90.6%.

To describe the value of different otoscopic findings in diagnosing MEE in clinically different cases a battery of variables are calculated in Table V. While redness of the TM was not very common (poor sensitivity) in cases with acute symptoms and MEE (AOM), its specificity figures were rather high. However, the incidence of false-positive and false-negative findings in regard to the absence and presence of MEE were unacceptably high, and the efficiency of redness to diagnose AOM low. For cloudiness all the variables showed rather good values, even in cases without acute symptoms. The same was true for impaired mobility of the TM, although the proportion of false-positive findings was rather high in Group II. Bulging showed poor sensitivity, very good specificity, few false-positive but rather many false-negative findings and not very high efficiency. The value of retraction of

TABLE V

Diagnostic value of tympanic membrane (TM) findings in diagnosing middle ear effusion (MEE)

TM-finding in visits	Sensitivity		Specificity		False-positive		False-negative		Efficiency	
	GI (%)	GII (%)	GI (%)	GII (%)	GI (%)	GII (%)	GI (%)	GII (%)	GI (%)	GII (%)
With acute symptoms										
red	17.5	26.9	84.5	84.0	40.4	48.6	56.1	35.3	46.5	62.0
distinctly *	13.6	24.4	92.3	92.0	30.2	34.4	55.1	34.0	47.7	65.9
cloudy	81.4	67.1	95.3	89.5	4.3	20.0	20.3	18.8	87.4	80.8
abnormal colour	98.9	94.0	79.8	73.5	13.5	31.0	1.7	4.9	90.6	81.4
bulging	61.2	41.3	96.8	96.9	4.0	11.0	33.9	27.1	76.8	75.7
retracted	6.5	19.4	90.6	87.9	53.2	50.4	57.0	36.0	43.3	61.8
abnormal position	67.7	60.7	87.3	84.8	12.7	29.0	32.1	22.2	76.3	75.6
impaired mobility	98.3	94.4	79.4	71.9	13.9	31.9	2.7	4.8	90.1	80.6
distinctly	82.6	84.0	93.1	85.4	6.0	21.5	19.5	10.6	87.3	84.9
slightly	15.7	10.3	86.3	86.6	40.3	67.2	55.9	39.7	46.5	56.9
Without acute symptoms										
red	4.5	12.3	94.7	91.9	75.7	77.0	27.0	15.7	70.5	78.8
distinctly *	3.3	11.3	97.1	96.6	70.5	60.7	26.7	15.2	72.0	82.5
cloudy	92.9	68.9	98.2	87.6	4.9	47.7	2.6	6.5	96.8	84.5
abnormal colour	97.4	81.1	93.0	79.6	16.5	56.1	1.0	4.5	94.2	79.8
bulging	45.2	18.4	99.2	99.2	4.3	18.6	16.8	13.4	84.6	86.5
retracted	10.0	32.1	94.9	90.4	58.1	61.4	25.8	12.7	75.4	81.2
abnormal position	55.3	50.5	94.2	89.6	22.4	52.2	14.8	9.4	83.7	83.5
impaired mobility	98.7	93.7	90.3	71.2	21.0	59.2	0.5	1.8	92.6	75.1
distinctly	76.3	84.3	97.5	85.8	8.5	44.2	8.2	3.7	91.7	85.6
slightly	22.4	9.4	93.0	85.3	7.0	14.7	77.6	90.6	74.0	72.1

G = group of children; see Materials and Methods.

* Haemorrhagic, strongly or moderately red.

TABLE VI

Likelihood ratios of the presence (LR +) and absence (LR -) of different tympanic membrane (TM) findings in visits with middle ear effusion (MEE) as compared to visits without MEE

TM-finding	Visits with acute symptoms				Visits without acute symptoms			
	Present (LR +)		Absent (LR -)		Present (LR +)		Absent (LR -)	
	G I	G II	G I	G II	G I	G II	G I	G II
Colour								
distinctly red *	1.8	1.9	0.9	0.9	0.5	3.0	1.0	0.9
slightly red	0.5	0.3	1.0	1.1	1.1	0.2	1.0	1.0
cloudy	17.3	6.4	0.2	0.4	51.6	5.6	0.07	0.4
normal	0.01	0.08	4.9	3.5	0.03	0.2	13.9	4.0
Position								
bulging	19.1	13.3	0.4	0.6	56.5	23.0	0.6	0.8
retracted	0.7	1.6	1.0	0.9	2.0	3.3	0.9	0.8
normal	0.4	0.5	5.3	4.0	0.5	0.6	9.5	4.9
Mobility								
distinctly impaired	12.0	5.8	0.2	0.2	30.5	5.9	0.2	0.2
slightly impaired	1.1	0.8	1.0	1.0	3.2	0.6	0.8	1.1
normal	0.02	0.08	4.8	3.4	0.01	0.09	10.2	3.3

LR +, LR -; see Materials and Methods.

G = Group of children; see Materials and Methods.

* Haemorrhagic, strongly or moderately red.

the TM in the diagnosis of AOM or non-acute otitis media was very poor with most variables measured.

The likelihood ratios of positive and negative otoscopic (= presence and absence of certain tympanic membrane) findings are presented in Table VI. Only cloudiness, bulging and distinct impairment of TM mobility revealed high ratios (LR +) in the presence of MEE. For redness of the TM the ratios were poor (low). Of the findings, cloudiness and distinct hypomobility of the TM showed also relatively good (low) ratios for the absence of the finding (LR -) in visits with MEE when compared to visits without MEE. In parallel, the posterior odds with different otoscopic findings in detecting MEE depicts the probability of finding MEE behind a cloudy, bulging or distinctly hypomobile TM, to be high, but that for TM redness to be low (Table VII).

There were statistically significant interobserver/intergroup variations in the incidences of the TM findings and in the variables derived from them. In Group II the TM was found to be more frequently red or retracted than in Group I (Table III). In cases with MEE, cloudy or bulging TMs were more frequent in Group I. If there was no MEE, cloudiness was more frequently recorded in Group II. In cases with MEE, mobility of the TM was usually limited, in cases without MEE, however, hypomobility was less rarely found in Group I. The specificities of different findings were about the same in both groups (Table V), but the predictive values of

TABLE VII

The odds (OPo) with different tympanic membrane (TM) findings in detecting middle ear effusion (MEE)

<i>TM-finding</i>	<i>Visits with acute symptoms</i>		<i>Visits without acute symptoms</i>	
	<i>G I</i>	<i>G II</i>	<i>G I</i>	<i>G II</i>
Colour	(1.31)	(0.63)	(0.37)	(0.20)
distinctly red *	2.3	1.9	0.4	0.7
slightly red	0.7	0.2	0.2	0.04
cloudy	22.5	4.0	19.4	1.1
normal	0.02	0.05	0.01	0.05
Position	(1.28)	(0.61)	(0.36)	(0.19)
bulging	24.3	8.1	22.0	4.4
retracted	0.9	1.0	0.7	0.6
normal	0.5	0.3	0.2	0.1
Mobility	(1.30)	(0.64)	(0.37)	(0.21)
distinctly impaired	15.6	3.7	10.7	1.3
slightly impaired	1.5	0.5	1.2	0.1
normal	0.03	0.05	0.005	0.02

OPo = posterior odds; see Materials and Methods.

G = group of children; see Materials and Methods.

* Haemorrhagic, strongly or moderately red.

In each TM-finding category prior odds, i.e. odds before otoscopy, are given in parentheses.

cloudiness, impaired mobility and bulging of the TM as an indicator of MEE were higher in Group I (Table IV). On the other hand, most of different false-negative TM findings were also more frequently recorded in Group I (Table V). The likelihood ratios of the presence of TM cloudiness, bulging, or distinct hypomobility in the presence of MEE (LR +) were higher in Group I. The respective findings also showed higher posterior odds in Group I, but only partly as a reflection of higher prior odds in that group.

Discussion

The most reliable way to identify the presence or absence of MEE is to perform a myringotomy. In this study it was always performed when MEE was suspected in pneumatic otoscopy, regardless of whether there were acute symptoms. When children had acute symptoms, MEE was not found in myringotomy in 15.1% and 18.2% of the cases in the two groups; and in cases without acute symptoms 30.9% and 31.2% of the myringotomized children did not show MEE in aspiration. Although it was not ethically acceptable to perform myringotomy in visits without a clinical suspicion of MEE, the remarkable proportions of myringotomies with negative aspiration suggest that probably almost all of the ears with MEE were found and myringotomized. Therefore, in the present study we can rather reliably

relate ear symptoms and otoscopic findings to the presence of MEE found in myringotomy and to the absence of MEE not found in myringotomy or even suspected in pneumatic otoscopy.

Myringotomy was more frequently performed by the otolaryngologist (Group I) than the pediatrician (Group II). However, the proportions of visits with positive and negative aspirations of all the visits with myringotomy were about the same with the two otoscopists. Similarly, the incidence rate of AOM was the same in the two populations. Thus, although there were differences in the frequencies of different otoscopic findings between the two otoscopists, and the two groups of children, the value of pneumatic otoscopy in general in diagnosing MEE was about the same in the hands of the two representatives of the two specialties taking care of children with ear problems. The relatively greater total number of visits per child in Oulu and greater proportion of visits with AOM of all the visits in Tampere suggest that the two study populations from geographically different areas, but with the same age distribution and AOM incidence, behaved differently as regards visiting the study clinics: in Oulu they visited more easily. However, the proportion of visits with acute symptoms was the same in both areas. This suggests that the visits classified as due to acute symptoms were in general less severe in Oulu, which may be one reason for differences in the incidences of different otoscopic findings between the two groups. Consequently, and because probably all clinical study materials are to some extent differing, no exactly true incidence figures for certain otoscopic findings, valid from one material to another, can be presented. Thus our figures must be taken only as approximates.

To determine the importance and diagnostic value of different otoscopic findings in the diagnostics of MEE, and especially of AOM, we calculated different variables from the incidence figures (Tables III–VII). None of them alone shows the real importance of a certain finding, but they all must be taken into account in assessing it. For instance, if the proportion of a certain finding is small in disease (low sensitivity), even rather high specificity figures do not indicate that the finding is of high diagnostic value. Furthermore, high predictive value of a certain finding may be good in indicating the presence of MEE, although the finding is poor for discriminating between cases with or without MEE. On the other hand, some derivate variables, like the likelihood ratio and odds as derivatives of sensitivity and specificity, suggesting here the ability of a certain test (finding) to discriminate between ears with and without MEE, may also leave the frequently occurring but benign nature of the disease unnoticed, and must be evaluated only together with other calculated parameters and variables. It is also to be noted that in this study we determined the value of single otoscopic findings, separately, in the diagnosis of MEE. However, in the clinical examination the otoscopist always registers a battery of findings (colour, mobility, position), and his decision on the existence or non-existence of MEE is based, in addition to the clinical symptoms, on the combination of these findings. An analysis of the clinical value of the different combinations of otoscopic findings will be presented separately [9].

The incidence of red TMs was surprisingly low even in AOM, only 21%. In acute visits without MEE it was 16%, and even in visits without acute symptoms 7%.

Thus, redness of the TM showed poor sensitivity in AOM. If present, it also predicted MEE (AOM) in only about half of the visits with acute symptoms. On the other hand, the great proportion of false-negative findings in visits with acute symptoms means that the absence of redness was a poor indicator of the absence of MEE, too. Redness of the TM also showed poor likelihood ratios (in general, the more $LR +$ exceeds 1 and the more $LR -$ is under 1, the better the discrimination ability of a positive and negative finding) in the presence of MEE and low odds for the presence of MEE even in acute visits. Thus, in accordance with some other reports [2,7,10,15], our present as well as earlier [13] findings suggest that redness of the TM is an inconsistent finding in AOM, and cannot be regarded as a reliable indicator of AOM. One reason for the great proportion of false-positive findings even in non-acute cases may be due to the TM redness caused by the crying and struggling of a child during examination.

Cloudiness (dullness, opacification) of the TM showed high sensitivity and specificity in diagnosing MEE even in cases without acute symptoms. Its predictive value was also high, with an acceptably low incidence of both false-positive and -negative findings. It showed also good likelihood ratios in and high odds for visits with MEE. So, it might be that a turbid, purulent and also mucoid MEE causes the cloudiness of the TM, with redness seen only in some specific phases, e.g. in the 'early' or 'hyperacute' cases of AOM.

Bulging of the TM is caused by positive pressure in the middle ear cavity, most often due to MEE. Bulging was found in only about one-half of the visits with AOM (Table III). On the other hand, in cases without MEE it was very rare and thus, if present, it very reliably predicted MEE. However, although the likelihood ratios and odds of bulging in regard to MEE were very good, the great proportion of false-negative findings weakens its overall value in diagnosing the MEE in acute visits (AOM). Nevertheless, its value in that respect was, although less than that of cloudiness, distinctly higher than that of redness of the TM. On the other hand, retraction of the TM was a very poor indicator of MEE with all the variables measured.

Impaired mobility was almost a rule in cases with MEE, 97% in acute as well as in non-acute cases, and its specificity for MEE was acceptable, 75% and 81%, respectively. Also the predictive values of the negative finding (normal mobility) for the absence of MEE were good. However, the value of distinctly impaired mobility (as well as of cloudiness) in predicting MEE seemed to depend on the otoscopist (Tables IV, VI, VII). In addition to pure interobserver variations, differences in the general clinical picture of the visits in the two groups classified into the same clinical categories (e.g. acute visits) might also have been causing these differences. On the other hand, cases classified as having only slight impairment of TM mobility seemed to be of no diagnostic value in either group.

Overall, in AOM and non-acute MEE the incidences of normal TM findings were very low except for position. The predictive value of the normal-looking (colour and/or mobility) TM for the absence of MEE was good, but various abnormal TM findings were of varying value in the diagnosis of MEE. Of the different TM findings, cloudiness, distinctly impaired mobility and bulging rather reliably indi-

cateú MEE, but redness, retraction and a slightly impaired mobility were poor predictors of MEE. As regards the presence or absence of these single findings in the presence of MEE, only cloudiness and distinctly impaired mobility can be regarded as having an acceptable discrimination ability, especially in acute cases. On the other hand, it should be remembered that none of the single otoscopic findings can indicate AOM but only the existence of MEE. For the diagnosis of AOM, in addition to the suggestive findings and combinations of findings, acute ear-related symptoms must also always be present.

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